Is

Relation Without Time, Cause, or Rule

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Draft — draft-2026-09-02.1
2 September 2026 • document revision 117

DRAFT — draft-2026-09-02.1 r117 (2 September 2026). This document is under active revision and is circulated for comment, not as a finished result. Definitions, numbering and claims may change between drafts; cite the version string above if referring to it. Sections marked as falsifiability commitments are the intended points of attack.



Figure 1: Is.

Abstract

One claim, made without machinery. To be is to stand in relation and to cohere in it, and nothing further is required for there to be anything at all — no sequence, no cause, no rule about what may follow what, *and no observer*. Observation is not primitive here, and it is not derived here either — it is *absent*, and the companions derive it later, once they have added what it takes. So what merely stands in relation cannot observe, and — nothing containing it — cannot be observed either, only modelled, and nothing in this account waits on anything looking. The companion accounts begin after this point and say so; this one supplies what they begin after. The claim is short. This paper is not, and on its own accounting need not be: what is being asked for is the sentence, and the rest is the case for it.

1 The claim

What is, is. That is the whole of it, and everything below is an attempt to say what the sentence commits to and what it does not.

One thing it does not commit to should be said before anything else, because the machinery below is routinely read as committing to its opposite. *Observation is not primitive, and it is not derived here.* It is *not present at all*: making a distinction is an act, acts need transformation and a grammar of what may follow, and this account has neither — so the deriving happens in the companions, at the point where they pay for those, and what this paper describes cannot observe — and, there being nothing to contain it, cannot be observed either, only modelled (Proposition 5.1). An observer is a late and expensive thing, and nothing here waits on one.

To be *something* rather than nothing at all is to be distinguishable, and to be distinguishable is to stand apart from something. *Distinction is relation.* There is no third option in which a thing is determinate and stands in no relation whatever.

That sentence is a commitment and not a result, and it is the one place to disagree. It holds that a determinate something must be distinguishable, and that distinguishability is irreducibly relational. Positions denying it are available and are not silly: a thing might be determinate

through an intrinsic character, a primitive thisness, or its own numerical identity, with no second thing anywhere. Nothing below derives this away. What is offered against it is a price rather than a refutation — such a character has to be *posited*, and it is a posit no relation registers, which is the one thing the norm in Section 2 charges for. A reader willing to pay it should stop at this line, because everything after it is downstream. *The disagreement belongs here and not several pages later.*

Three claims come apart, and the paper is easier to attack once they are separated. *One:* determinacy is relational — the commitment just made. *Two:* time, cause, rule and an observer are not entailed by it, which is what the removals below establish. *Three:* removal against a shortest-sufficient standard is a reasonable way to audit what an account carries, which is methodological and neither of the first two. The second and third can fail without touching the first, and the first can be rejected on its own.

Proposition 1.1 (To be is to be related)**.** *Anything that exists as a determinate something stands in at least one relation. Contrapositively: where no relation holds, there is no distinction, and where there is no distinction there is no determinate something — not one that cannot be told apart from nothing, but no second thing there to be told apart.*

Figure 1 is the claim. There is nothing else to draw, and the omissions are the content: no names, no types, no arrowhead, no distinction between the marks beyond where they sit, and *no border* — a border is a boundary, a boundary belongs to whatever draws it, and at this level there is nothing to draw one, nor anything for the figure to be inside. It demonstrates what it states rather than illustrating it. What makes the marks two is their position on the page and nothing in them; a reader taking the left one as first reads a direction off the paper. The page is the container the drawing cannot escape and declines to assert. All of that is supplied. What is not is that there is a difference here at all.

What is is the subject, and nothing else is. Where a sentence here looks as though it quantifies over what does not exist, the failing item is a description and not a thing — the care the first proposition takes in saying *no second thing there to be told apart.*

What the words carry

Four words do the work, and each is used in one sense throughout. They are given here rather than assembled as they arrive, because a reader who lets any of them drift between an ontological, a logical and an informational reading will find the propositions below proving more than they do.

Relatum. An end of a relation, and nothing else. Not a substance, not a bearer of intrinsic properties, not an object carrying an identity condition supplied from somewhere — there is nowhere here for one to be supplied from. What makes two relata two is the subject of Section 3 and is not assumed here.

Relation. A tie with ends. Not an ordered pair drawn from a domain fixed in advance, which is the textbook construction and is a *construction*; why that order is unavailable at this level is given in Section 10. It carries no type, no direction and no weight, each of those being supplied by whatever describes it rather than found in it.

Distinction. Two ends failing to collapse into one. This is the identification made above, and it is where disagreement with this paper belongs.

Coherence. A relation coheres when its ends are distinct. That is stated as a condition and not derived from anything, and saying so is the honest version: it is the structural exclusion of the two zero-distance cases — nothing related, and a thing related only to itself — and the exclusion is what does the work everywhere below. It is not an epistemic measure, presupposes no observer, and requires no informational state, language, frame or evaluator, none of which exist at this level and all of which the account would have to buy. The informational reading — that a relation carries a fact about one end given the other that was not already had — *agrees*

with it here, and is how the companions carry the criterion forward once there are frames to measure against. The agreement runs in that direction only. *The base condition is structural; the informational criterion is what it becomes once there is something to measure with*, and a reader who takes the second for the ground of the first will look for an evaluator this level cannot supply and correctly fail to find one.

One consequence of the last is worth stating rather than leaving to be inferred. A criterion for existing that *needed* a frame would index existing to frames, which is the reading this account refuses everywhere else; so the floor has to be the structural one. *What carries nothing carries nothing for anything.*

2 Remove, and see what breaks

A claim of this shape is worth nothing if it arrives by assertion, and the difference between an account and a pronouncement is whether there is a procedure behind it that someone else can run. There is one here, and it is short.

One thing about it has to be said before the norms, because a reader will otherwise supply the wrong object. What the procedure removes are *candidate posits in an account* — time, cause, rule, substance, an observer — and not edges from a graph. *The graph is what the procedure arrives at, not what it operates on*, and nothing below is a claim about what breaks when a structure is cut. The vocabulary that follows is accordingly the vocabulary of assessing accounts, and it is used here and nowhere else in this paper.

That is one instance of a distinction the whole paper rests on, and it is cheaper to state once than to rely on. The base has no logic, no inference, no norms and no rules: that is the claim. Stating the claim uses all four. There is no contradiction in this, and there would be one in pretending otherwise — an argument is conducted in a metalanguage, and what a metalanguage requires is not thereby part of what it describes. Inference, the exclusion of incoherent descriptions, the preference for the shorter account, the removal tests, the judgement that coverage held: every one of these belongs to the assessing and none to the assessed. *A description of a level with no rules is not itself at that level.* The reader is owed this because without it the denial of rules reads as self-undermining, which it is not; and where this paper slips and attributes a metalinguistic resource to the base, that is a defect and should be reported as one rather than absorbed.

Ask at maximum scope. The question is put to everything at once rather than to a domain, because a shortest sufficient account of one domain tells you about that domain and a shortest sufficient account of all of it tells you what is common to all of it. The scope is what does the work: the more that must be accounted for, the more the structure peculiar to any part washes out and the more only what is shared survives.

Prefer the shortest account that suffices. Between two accounts covering the same ground, the one carrying less structure is preferred, and this is the ordinary discipline of minimum description length rather than anything invented here. Applied at maximum scope it has a direction: keep removing until removing costs coverage.

It is also why *shortest* needs a frame, and that is not a defect: description length is defined only against a vocabulary, which is what the companion's Axiom −1 insists on.

Remove, and see what breaks. This is the operation and it is the only one. Take a candidate — time, cause, rule, substance, an observer, a global clock — and put the account without it. If what was accounted for is still accounted for, the candidate was not required and it goes. If coverage is lost, it stays, and the loss shows what it was doing. Each removal is a test whose outcome is determinate in one direction and not in the other, and the asymmetry is this: a *successful* removal settles the matter, since an account that stands without the candidate is exhibited and can be inspected, while a *failed* one records only that nobody has yet seen how, and stays open to a reader who does. That is what makes the procedure repeatable rather than

a matter of taste.

Run that far enough and it stops in one place. Time can go: nothing in the claim requires sequence. Cause can go: relation is not production. Rule can go: nothing needs to be permitted or forbidden for things to stand as they stand. Relation cannot go, because removing it removes distinction, and removing distinction removes the possibility of there being determinate somethings for the account to be about at all. *The claim is not asserted here. It is what survived* — survived, that is, given the identification made in Section 1. Relation is what cannot go *because* distinction was granted to be relational; the removals show that nothing else is needed alongside it. They do not establish the identification, and nothing in this paper does.

A removal must be named and restorable. This is the discipline that separates the method from mere simplification, and it is the one place the procedure can be abused. Every account reached this way is an idealisation, and an idealisation is just a frame — which is no defence on its own, since everything is one. What makes a frame cheap to hold rather than expensive is whether what it removed is written down and can be put back. The rock is what this claim looks like at the grain at which it is a rock; that its atoms move and it weathers at other grains belongs on the same page, so the removed structure is recoverable and the idealisation costs little. *The spherical cow is not objectionable for being spherical.* It is objectionable when nobody writes down that it was made spherical, and the result is then applied to a cow.

Which this paper owes of itself, and this is the place. *It was made spherical by removing frames.* Nothing described here is individuated by anything, and everything real is: individuation is performed by a container, that container is individuated by its own, and there is no last one. So the base set out here is not a thing anybody meets, and it cannot be reached by stripping either, the stripping not terminating. *Nor is it an earlier stage of anything.* What is described is an idealised ontological floor and not a temporally prior physical state — there is no time here for it to be prior in, and a reader supplying a cosmological reading is restoring the sequence this section exists to remove. Written down, then: what was removed is frames, acts, and any grammar of what may follow, and where it goes back is the companions, which restore all three and pay for them in a ledger.

The kind of removal matters and is part of writing it down. A spherical cow is *false* of a cow. Nothing here is false of anything real — every actual thing does stand in relation and does cohere, and goes on doing both once its frames are put back. This paper is true of everything and complete about nothing, which is the only shape an account at this scope could have. *It is an idealisation by subtraction and not by distortion*, and the subtraction is the whole of what makes it short.

It also has to come first: a base that already assumed frames would be assuming the thing the companions exist to pay for, and their pages would then read as commitment rather than as derivation. Starting from an object with no instances is the cost of starting anywhere at all.

The norm is not this programme's invention. It is Occam's razor, and Rissanen (1978) is where it stops being a preference — *do not introduce structure that buys no additional distinction* — since *simpler* settles nothing until someone says simpler in what units. Shannon (1948) supplies that measure and *not* the norm, containing no model selection and treating redundancy as useful on a noisy channel, so citing the two together for the norm borrows a name that did not make the claim. What is added here is only the scope at which it is applied.

Removal establishes that nothing *tried* was required, which is not the same as nothing being required. The result is only as good as the candidate list, so that list should be stated rather than implied: time, causation, substance or intrinsic nature, an observer, a privileged frame or global clock, admissibility, and — less obviously — any axiom restricting where a distinction may be made from.

The last of those is worth naming, because it is the one kept by habit: a rule that every distinction is made in some frame governs frames, and there are none here to govern. It arrives with the machinery it constrains, in the companion account, and is not required for anything

this paper says.

Removing it does not remove every commitment about distinction, and the two are easily run together. *Removed*: any rule about where a distinction may be made *from*. *Retained, and load-bearing*: the denial of a privileged characterisation, which forces the graph above and is this paper's one prohibition; and that a distinction is a relation, which is the identification the whole claim turns on, and a commitment rather than a discovery. A reader who holds that a thing can be determinate without standing apart from anything rejects this paper at that sentence and needs to go no further — which is where the disagreement should be, rather than several pages downstream.

Every entry on that list is a *posit*, something an account might assert exists, and a list of one kind is short by a whole category. A commitment can also arrive as a *word*: vocabulary carrying something it does not declare, which no reader stops on because it does not present as a claim at all. *Type* is the worked case — calling a graph typed asserts that kinds are part of what is there, and the companion settles that they are not, distinctness being forced where sameness is supplied. *Substance* is worse, being already on the list and returned anyway in a companion's own sentence about having decomposed it away. The operation is the same one; only the candidates are new. Remove the word and see whether any claim alters: vocabulary doing no work goes silently, and vocabulary carrying a smuggled commitment takes a claim with it.

Anyone who thinks something else belongs at the base is invited to add it and run the same procedure, and that is the form disagreement with this paper should take.

One operation, applied to what came bundled

The operation is stated above as removing posited structure — time, cause, rule — but its more productive target is different, and is the same operation pointed the other way.

Almost every claim worth taking at maximum scope is one that some field already accepts, and accepts with a restriction attached. Relations are basic *in this formalism*. There is no privileged observer *in mechanics*. The shortest description is preferred *for this class of model*. The restriction is generally not a discovered property of the subject; it is the restriction that constitutes the field, and it travelled with the claim because within a field there was never an occasion to separate them. So the move is to take the claim seriously and decline the restriction — which is not a new procedure but *remove, and see what breaks* applied to an inherited limit rather than to a posited entity, with the same asymmetric outcome and the same requirement that the removal be named and restorable.

Two things follow. The first is that results reached this way look simultaneously trivial and consequential, and are both: trivial because nothing was added and the claim was already granted, consequential because the discarded restriction was doing work that nobody had audited, having never been separable from the subject inside the field that held it.

The second is a limit, and it is severe. Most restrictions are load-bearing — they are attached because removing them breaks something, and taking a claim past its restriction usually produces nonsense rather than generality. The operation is cheap to run and returns nothing on almost every attempt — a rate this paper commits to in Section 11 and cannot yet support, since no ledger of the failures was kept and the base rate is therefore reconstructed rather than recorded — and the ones that return something are worth reporting for that reason and not because the operation is powerful. Nothing here is a recipe for producing results by taking things literally. It is a cheap test, run often, with a low yield, and a signal on firing that is clear about *that* something did not break while saying nothing about whether what survived is worth having.

3 One is not enough

A single relatum is a valid candidate for examination and this paper does not get to forbid it. One may formulate it, draw it, name it and investigate it — all of that from a frame outside the case, which is where the formulating happens and is exactly the metatheoretic position Section 2 sets apart. *What it has is nowhere to be asked from.* Asking-from requires a frame that can make and hold a distinction; a lone relatum has no second end, no boundary, no relation, and therefore no internally available distinction — not even, internally, from the absence of the second relatum whose absence is the whole of its condition.

So the verdict is not that the one-node case is meaningless or that it cannot be drawn. It is that on the commitment of Section 1 it does not constitute a determinate existent: it supplies no relation by which its relatum could be distinct from anything. *One relatum does not cohere*, in the sense the definition above gives — its ends, such as they are, are not distinct. It does not sit at the bottom of the hierarchy of existing things; it fails to enter. And it earns its place here by being the candidate that shows why the minimum must contain a relation with *distinct* ends rather than merely a relation: not one node, but two relata in a non-self relation.

The empty case is the other end and is cleaner. Nothing at all stands in no relation, has no parts to be in tension, and admits no failure of fit — it is complete in the only sense available to it. That is why the empty set is the only perfect thing and why perfection is not something an existent can have: *to exist is to be imperfect*, because to exist is to be related, and to be related is to be partial with respect to what one is related to.

The two ends meet, and where they meet is the point. The failed case yields nothing because the relation has no distance to cross; the empty case yields nothing because there is nothing to cross to. Reading either off gives the same result — nothing — and so does a third: everything at once, read from outside, has nothing left over to stand in relation to it, and gives nothing too. *That is the correct reading of the old claim that everything and nothing come to the same*, and it is a claim about two *readings*, never about two things. The version that drops the qualifier asserts the identity from a standpoint, and there is no standpoint at which it holds. From anywhere at all — which is the only place anything is — everything is emphatically not nothing. Between the two zeros is where the distinctions live, and everything this paper is about lies there.

4 Coherence, and what fails to cohere

Proposition 4.1 (To exist is to cohere)**.** *A relation suffices only where it* coheres*: where what it reports about one end given the other was not already given. Neither degenerate case is a thing that fails the condition: where no relation holds there is nothing to cohere, and where the only relation is to oneself there is a relation and nothing carried by it. In both, what fails is the* description*, which picks out nothing. Relation is necessary; coherence is what makes it sufficient.*

The second proposition is not a refinement offered for tidiness. The first, read alone, is satisfied by a thing related only to *itself* — the letter met and the point defeated, since a relation to oneself reports nothing that was not already given. That is why the prose above says *distinguishable* where the first proposition says *stands in at least one relation*: the two come apart there, and the prose was the accurate one. To cohere is for a relation to *carry* something — a fact about one end given the other that was not already had — which makes the criterion a quantity rather than a condition met or not met by inspection, and sorts every case by that one measure. Nothing related: no relation, nothing carried. Related only to itself: a relation, nothing carried. Two things related: something carried, and the only one of the three that is anything. *Things that do not cohere are not there to fail.*

Two further readings sort the same three cases the same way, and neither grounds the condition. Consistency asks whether things can be held together; information asks whether

holding one tells you about another. Read either way the cases sort identically, and a self-relation lands at the end where holding is trivial rather than the end where it is impossible — which is why its failure is emptiness and not prohibition. The agreement is worth having, being what lets the companions carry the criterion forward as a measured quantity. It is not an argument for the condition, and this paper does not offer it as one. The verdict is settled without a frame, for the reason the definition of *coherence* above gives, so the degenerate cases are degenerate absolutely.

It is tempting to conclude that rules are needed after all — that something must *enforce* coherence, and enforcing is what rules do. The temptation comes from hearing the question as *what can be created*, and the question here is *what can exist*. Creating needs an agent, an act, and a grammar governing the act, and this paper has none of the three. *Nothing excludes the incoherent cases, and nothing needs to*, not existing being no outcome that has to be brought about. What coheres, coheres; what does not is not there; and no rule stands between the two doing work.

The smallest case shows the two coming apart. A graph of one node can be *created* — drawn, named, saved to a file — and does not thereby exist in this paper's sense, one relatum standing apart from nothing. Labelling it does not rescue it, a label being supplied by whatever draws it rather than a relation the node has, which is the point Figure 1 makes about its two marks. So creation is neither necessary nor sufficient for existence here, and the companions — about worlds in which things happen, and therefore about making — are where rules belong, as is something already picked out ceasing to cohere.

5 One direction only

A consequence of the restraint deserves its own statement, because it is what makes the claim's subject matter peculiar. *What merely is cannot observe.* Whether it can be *observed* turns on whether anything contains it; in the case posited here nothing does, and it can only be *modelled*.

To observe is to make a distinction, and making one is an act — something is done, and afterwards the observer holds what it did not hold before. That requires transformation and some grammar of what may follow what, and this paper has neither. So what is described here does nothing at all, and observing is among the things it does not do.

Two cases have to be separated here, and running them together is an error. *Where the base is embedded* — a level inside a world that has containers — it can be the object of observation without difficulty, and a rock is what that looks like from outside: met, related to, distinguished by something that does have the machinery.

The scenario posited here is the other case. The base is embedded in nothing, because it is everything there is. *So there is no container to instantiate a boundary* — and instantiating is what a container does, as this section has just said — *so there is nothing to observe it, and no observation occurs.* To say it *can be* observed grants an outside the scenario does not contain, which is the free import of a relation, in the section whose whole work is to refuse exactly that.

What remains available is modelling. The model is a metatheoretic representation *of* the base — of it, and not merely of the modeller's own notes, since a representation of nothing in particular would be arbitrary rather than non-observational. What it is not is an observation performed inside the scenario: the representing relation belongs to the modeller's frame and adds no edge, no boundary and no observer to what is described. *One can model what one cannot touch*, and touching is what is unavailable here. *Which is what this paper does.* It does not observe what it describes. It models it, from a frame that is not in the scenario at all — and saying so is not a confession but a statement of the only access there is.

But there is no corresponding inside. *There is no view from anywhere* — not a narrow view, not a limited one, none.

The reason is not that framehood needs rules — it does not, and the companion is explicit that a structure with no rules of its own is still a frame, because *individuation is performed by*

the container and not by the thing. A boundary belongs to the frame that draws it, so what a rule-less structure lacks is not framehood but the capacity to individuate anything itself. It is also why an *empty* frame is impossible rather than merely unattested: an edge needs relata, so where there is nothing there is nothing to demarcate.

What the base lacks is the drawing. A frame is a boundary relation *instantiated*, and instantiating is something a container does; a level at which nothing runs has nothing doing it, so it has no frames — not because a frame would need rules of its own, but because it needs someone else's.

What is missing is *making.* A view is not had by standing in a relation; it is had by a frame that *makes* a distinction, and making is doing, and doing is what this level does not have — that is precisely what the companions add when they admit that things happen. So the familiar claim, that every view is had from somewhere by someone positioned, presupposes there are views and asks where they are from. Here there are none to ask about: *no view from nowhere, and no view from anywhere either*, not for want of a place to view from but for want of anything doing the viewing.

An axiom restricting where a distinction may be *made* from therefore cannot be stated at this level, let alone assumed — because it governs an act, and there are no acts here — which is also why there are no frames at this level to govern, since instantiating a boundary is one. The other half of that axiom is a different matter and is retained: denying that any characterisation is privileged says nothing about acts or frames, and it is what makes the graph a result rather than a preference. It arrives with the machinery that performs them.

Proposition 5.1 (Observation is one-directional at the base)**.** *What stands in relation without transformation or grammar cannot observe. Where a container exists it can be observed, and is then always the object of a distinction and never the maker of one. Where none exists — the case posited here —* it can be modelled and not observed*, modelling requiring no relation to its object.*

6 Which is to say: a graph

The claim has a name, and refusing to use it would be coy. Relata standing in relations are nodes standing in edges. *Everything is a graph* — and that sentence is the whole thesis of the companion accounts, stated here in the only form that requires nothing to state it.

Which graph has to be said, because the word carries a construction that is not meant, and a reader supplying the textbook one will find the claim presupposing most of what it denies. At the floor meant here, *graph* names no labels, values, types, directions, weights, metrics or external embedding. It names relata and a relation between them, and nothing else. A node is an *end* of a relation, recoverable from the relation rather than prior to it — which is the definition given above and not a second one, and it is why the two nodes of K_2 are nodes although neither is where anything meets — the reverse of the order mathematics needs, for reasons Section 10 gives in full and which are not a quibble about notation.

Three things that follow are worth settling here rather than leaving to be reconstructed.

The minimum is not an arbitrary graph with an edge in it. It is two nodes joined by one edge. A lone node fails for the reason Section 3 gives, and a self-loop fails for the reason Proposition 4.1 gives: each supplies a structure without supplying a nontrivial distinction, and neither is a small existent but a failed one. So the smallest coherent instance is K_2, and the condition on a base graph is not that its edge set is non-empty but that some edge has distinct ends.

The two ends are distinct without being qualitatively different, and this is the sharpest place to press the argument. In K_2 the nodes have identical structural profiles: swapping them changes nothing, and there is no label, intrinsic mark or property to tell them apart — by construction, since the floor has none to offer. What makes them two is the irreflexivity of the relation between

them. They are *weakly discernible*: distinct as the two ends of a relation that nothing bears to itself, which is discernibility without a discerning property and is exactly the notion the identity of indiscernibles was found to need (Black, 1952; Saunders, 2003). That literature reached it for physics, where the case is fermions rather than a base graph; the notion is the same one and is not this paper's invention. A reader who rejects weak discernibility rejects the second node here, and lands back at the commitment in Section 1 — which is the correct place for the objection to arrive.

The set-theoretic apparatus is the metatheory's encoding and not additional furniture. Sets, ordered and unordered pairs, membership and the incidence construction are how an account writes a graph down; the ontology asserted is nodes and an edge. This is the object-language distinction of Section 2 applied at the point where it is most easily lost, and it is the same distinction that keeps "no rules at the base" consistent with the formal rules used to state the claim.

That step needs one thing, without which the graph is a choice of notation among richer ones and the obvious question — why not a manifold, why not something with an order or a metric built in? — has no answer. One answer suggests itself, is not available, and should be refused first, because it is the one this paper would otherwise be read as giving. *A richer intrinsic form could be asserted only from a standpoint at which the extra structure is visible, and no such standpoint is available* — from which nothing follows about whether the form is there. What can be asserted and what is the case are different questions, and an argument running from the first to the second is the slide from an epistemic limit to an ontic absence. The prohibition below constrains what may be *claimed*. It does not by itself empty the world of whatever no claim could reach, and this paper does not get to have it both ways.

What does the work is the measure, and it is already on the table. Structure making no difference to any relation buys no additional distinction, and the norm of Section 2 — Rissanen's, not this paper's — removes precisely that. A metric earns its place wherever some relation registers it; where none does, it is carried and never read, and carrying it is the silent cost the whole method exists to charge for. So the account declines it. *That is a claim about the account and not a proof about the world*, and the difference is not cosmetic: someone willing to posit an intrinsic form that no relation registers may have one, and pays a posit for something no relation could ever report back. This paper's method is the refusal to pay on those terms. It is weaker than a demonstration of impossibility, and it is what is available.

The denial is sharper as a measure, since the phrase invites the wrong picture. A view is informative in proportion to the relation between viewer and viewed — what the one tells you about the other is what a view delivers. A view from nowhere stands in no relation to what it views, so it delivers *nothing*: not a partial view or a distorted one, but zero. So it is not the complete view we are being denied; it is the empty one, and a standpoint that adds nothing to what it sees receives nothing from it either. Read off from nowhere is read off at no information, which is not reading anything off — and that is why what survives is relation and nothing above it.

Which means this paper is not adding an axiom here, and the companion's numbering of it as *Axiom* −1 should be read accordingly. To be is to stand in relation; a relation is what carries information; so a standpoint in no relation to what it views carries nothing about it. The denial follows from the claim and the measure, and posits nothing further. It is stated all the same, and stated as a prohibition, because what it denies is imported for free — a reader supplies the standpoint without deciding to, so a consequence nobody would otherwise draw has to be put where the import happens. So what survives every standpoint is the bare relational minimum, and the bare relational minimum is a graph. The graph is not selected for convenience; it is what is left when no vantage is allowed to add anything.

This is the companion's Axiom −1 doing a job at this level, and only half of it: the frame-governing half has nothing to govern here, for the reason Section 5 gives. What is used is the

denial of a privileged characterisation, and it is used as a constraint on claims rather than as the ontological premise the previous paragraph declined to make of it.

It also gives this paper the one prohibition it has, and why it has to be a prohibition rather than another removal explains the companion's odd numbering. Everything else here is removal: take a candidate away and see whether the account still stands. This one cannot be removed that way, because *it is not in the text to be taken out*.

A reader does not arrive neutral on whether there is a view from nowhere — they arrive already using one, having read a claim as being about how things are rather than how things are from somewhere, which is the default and costs nothing to adopt. What is imported for free returns unless it is refused explicitly, and an axiom that cancels a default has to be stated before the counting starts. Hence a prohibition, and hence its number. A world in which some structure has an intrinsic form readable from no standpoint would violate it.

The question this paper asks is the dumbest one available: *what is there?* A child can ask it and nothing prevents anyone answering. It is also the largest, because an answer to it constrains every answer to anything else, and because there is nowhere further back to stand.

The form of this paper follows from that. It builds *no machinery*, deliberately: no transformation, no grammar, no economics — none of the apparatus a reader could dispute *instead of* disputing the claim. That is not the same as offering nothing else to disagree with. What follows below is a method, a reading of how formalisms move under compression, and an accounting of what derivation costs; each is separable from the claim and each can be rejected on its own. The form that survives is narrower: no *structure is posited* here that could absorb an objection meant for the claim, so a reader who rejects the claim cannot be answered by adjusting a mechanism, because there is none to adjust.

The objection that arrives immediately can be answered in a line. *Everything can be represented as a graph, so the claim is empty* — yes to the first clause, and this is not a claim about representability. Nodes and edges are cheap; what is asserted is that there is *nothing else*, that relata and relations exhaust what there is, and that time and cause and rule in particular are not among the furniture. What would bear on it is a demonstration that this world has time or cause in its floor — *which is empirical and not modal*, so imagining a world that does costs nothing and settles nothing.

Finding one would bound the account rather than refute it, in the way a later theory bounds an earlier: the claim would hold of worlds without such a floor and this would not be one of them.

7 A consequence with a track record

The claim fixes what compression is aiming at, and that turns out to be checkable in a way the claim itself is not.

If what there is, is relata standing in relations, then the shortest sufficient description of anything is stated *in terms of* relata and relations. The qualification is doing real work. Without it the claim becomes that a description shorter than its subject's structure has lost something — which is a picture theory, and worse, contradicts the accounting two sections above. Minimum description length charges for a generator, not for what unrolling it produces: the shortest sufficient description of the complete graph on a million vertices is a phrase naming the relation and the count, not five hundred billion enumerated pairs, and a description isomorphic to its subject is very often the expensive one. So the claim cannot be that the shortest description *enumerates* relata and relations.

What it can be, and is, is a claim about the vocabulary the generator has to be written in. "The complete graph on a million vertices" is a short generator whose terms are a relation type, an arity and a count; it is about relata and relations without listing any. *The arity and the count are the generator's parameters and not values borne by anything at the floor*, which is

worth saying here rather than leaving to Section 11 to concede: they belong to the description doing the generating, in the metatheory, and the thing generated carries no labels at all. A reader who reads them as furniture has found the objection that section takes up, and has found it one section early. The prediction is therefore that formalisms under compression converge on *primitives* that are relata and relations, not on descriptions the size of their subjects. *The ideal information shape is a graph* in that sense — what the terms denote is fixed by what is being described, while how briefly they can be combined is not. So a field pushing its own formalism toward minimum description should *move toward* graph form, and it should do so without anyone intending it. *That is the older claim that information is what everything is made of* (Wheeler, 1990; Bateson, 1972), stated for everything rather than for physics.

That is a prediction about the history of formalisms rather than about the world, and it appears to hold: physics reaching for spin networks, tensor networks and causal sets; chemistry settling on the molecular graph; linguistics on dependency structure; computing on data structures that are graphs and on a categorical foundation in which the morphisms are primitive and the objects recoverable from their identities — which is the vertices-from-edges order this paper needs, reached there for reasons of its own; neuroscience on the connectome; biology on phylogenies and interaction networks. None of these was adopted because anybody held the thesis of this paper. Each was adopted because it was the cheapest thing that still accounted for the subject.

One case is set out rather than listed, being the oldest, the longest kept, and it is usually described as an invention. *Double-entry bookkeeping is not an invention. It is a compression.* A transfer has two sides, and the record that keeps both of them names two accounts and the movement between them — relata and a relation and nothing else. It should be said at once that this is not a minimality claim, because Ellerman refutes the obvious one: a single-sided system using signed numbers records the same information, and given the same opening equation and the same transactions the two end in the same place. What single-sided recording loses is the trial balance, and that is the whole of the difference worth having here. Nobody derived it from a thesis about what exists.

The history is what makes it a case rather than an analogy. Pacioli (1494) is usually credited with the system, and what he wrote was a description: the *Particularis de Computis et Scripturis*, a treatise inside a mathematics textbook, setting out a method already in use among Venetian merchants. The practice is older than the account of it by roughly two centuries — Lee (1977) reads the Farolfi ledger of 1299–1300 as full double entry — so the system was found in use and then written down, which is the order a compression arrives in and not the order an invention does. And it has been kept for five centuries, in the formalism the field teaches and computes with, which is what the commitment below actually asks for.

The counterweight belongs here too, because the history is contested in a way that bears on the claim. Yamey (1949) argued against the then-standard view that double entry was adopted for rational profit calculation, holding that early merchants did not use it that way. What Yamey found is a *smaller* purpose rather than none — tracking debtor and creditor relations, custody, and agency accountability — and Sangster (2016) names it more precisely: the system gave a clear picture of every debtor and creditor *together with the means to check that the entries between them were complete and accurate.* So Yamey does not license the inference that the method was adopted with no theory of what it was for. It was adopted for something, and the something was checking — which forecloses any argument here from nobody having designed the technique for anything, and means the error-detection property below was not a lucky side effect but what the thing was built to do, and why it was kept.

And it earns the case rather than illustrating it, because of what the practice detects. A ledger that does not balance is not a rule violation but a *residual* — something recorded that does not cohere with the rest — and the trial balance is a coherence check, which is why the technique catches errors at all. A convention that did not match the two-sidedness of what it

records would have no such property, so the survival of the technique is evidence about the subject and not only about the bookkeepers.

That reading has a mathematical statement and it is not this paper's. Ellerman (1985, 2014) shows that double entry implicitly uses the *group of differences* over pairs of unsigned numbers — the T-account is the pair — so a transaction is a difference and the two entries are its two components rather than one record written twice. Balance is then the group identity, and an unbalanced ledger is a residual in the exact sense, which is where the error-detecting property comes from. The dating is the part that bears on the claim here: the structure was in use long before anyone had the mathematics to say what it was, and the interval is given exactly below.

And Arya et al. (2000b) reach the relata-and-relations reading without the thesis, characterising the consistent transaction vectors by a *directed graph representation of the accounting system*, without which recovering transactions from statements (Arya et al., 2000a) becomes intractable as accounts multiply.

The discriminator matters and is easy to lose. *The prediction is about movement, not about drawability.* That a field's subject matter can be drawn as a graph shows nothing, since graphs are expressive and almost anything can be. What would count is a field that compressed its formalism and moved *away* from graph form — toward a primitive that is not a relatum or a relation and cannot be reduced to them — and that is what the second commitment below asks for.

8 What is not claimed

The claim above is smaller than it may appear, and the restraint is the point.

No time. Nothing here requires sequence. Relations hold; they do not happen. There is no earlier and no later in this paper, and no order in which anything is arrived at.

No cause. Nothing here makes anything else be. Relation is not production, and the account has no resources with which to say that one relatum brings about another.

No rule. Nothing here forbids anything or licenses anything. There is no admissibility, nothing that may or may not follow, and therefore nothing that could be violated. The coherence criterion does not smuggle one back, because *nothing enforces it*: what coheres is not permitted to, it simply does.

Those are not oversights and they are not deferred out of modesty. *They cost extra* — and the third is what *making* anything requires, which is why a paper about what can exist rather than what can be created does none of it. A world in which things also *happen* requires transformation and a grammar of what may follow what, and that is what the companion accounts introduce and pay for. This paper is what remains when they are removed, and it is offered as evidence that they are separable rather than constitutive.

9 What the companions add

The companion accounts begin from this claim and add to it what a world in which things *happen* costs, which is not required for there to be anything.

And they begin with a graph that is not an axiom of theirs. This paper does not assume that everything is a graph. It establishes that, given relational determinacy and the no-unpaid-structure discipline, *the bare untyped graph is the least ontology this account may retain.* The qualification is not a hedge appended to a stronger claim; it is the whole of what Section 6 licenses. So the companions inherit graphhood as a *result within that discipline* rather than as an independent postulate, and their axioms concern *change* in that graph rather than the graph's existence. What is established nowhere here is that nonrelational structure registering in no relation is impossible. Someone may posit it. What this account says of it is that it is unwarranted, which is a different word from unavailable and is the only one earned. The

companion's own ledger, corrected after audit and moved in both directions since, carries the admission and two postulates named there — a realisation operator and a destroy/preserve distinction — with its Axiom −1 a discipline against importing unpaid assumptions in that paper and a commitment that can fail in the groundings.

Three things are added and none of them is free: transformation, so that something does the happening; an economics, so that a change has a price; and realisation, so that what is admissible becomes what occurs. *The case that each is cheap is made in the companions and not here*, which is where it belongs — this section says what is taken on, not what it costs to take.

So what the companions add to this paper is the admission that things happen, two postulates named in their ledger, and past those the bets they state as hypotheses rather than as commitments. What those commit to together is then unpacked at length, and the unpacking is not further claims about what exists.

The strong form of the statement is therefore false and the useful form survives: the base is small, what is added to it is enumerable, and anything in the companions that is an independent posit uncounted by that enumeration is a defect in their economy. That standard has been applied to them and has moved the count more than once; the companions do not exempt themselves from it, and the record of its application is in the companion on assessment rather than here. Their not exempting themselves is the prohibition above applied to the writing: an account stating its rules from outside its own scope would be speaking from the standpoint this paper denies, and that standpoint is imported freely enough that refusing it once is not enough.

10 Prior positions

None of this is new and it should not be dressed as though it were. The neighbours are easy to name. That relations are prior to, or on a level with, their relata is the ontic structural realist position (Ladyman and Ross, 2007; Esfeld, 2004); that a thing is nothing over and above what it stands in is older and starker (Nāgārjuna, 1995); and the bundle tradition reaches a neighbouring conclusion by a different route (Williams, 1953).

The relation to them is not the usual one, and the difference can be stated exactly rather than as a matter of emphasis. *Each of those accounts supposes a particular* ***type*** *of relation where the argument only requires that there be one.* Ontic structural realism is a thesis about physics — about what our best theories say the world is made of — and its restriction is that *domain* rather than any particular edge. The structure it argues for is physical structure, and its live disputes are about objects, relations, modality and identity; characterising it as an observer edge would be a misreading, and this paper does not need one. The perspective literature is the case where an edge genuinely is fixed, viewing being constitutive there. Neither is mistaken about the structure. One is restricted to a domain and the other to an edge, and the removal applies to each in its own way.

The removal is immediate and is the method's own. Where an edge is fixed, take it out and put any other in its place — a gravitational one, a chemical bond, mere adjacency, one part of a rock against another — and everything the structural claim needs is still there. Two relata and a relation between them do not care which relation it is, so what is treated as constitutive is a *choice of edge type*, salient in the domain it was written for and doing no work in the claim itself. Where a domain is fixed instead, the removal is the other one: declining an inherited restriction, asking of everything what physics licensed about physical structure, and seeing whether anything breaks. In both cases what survives is the claim this paper makes, and in neither case is that claim new.

The same over-specification is visible in the two fields that own this vocabulary, which are the ones a reader will expect to have settled the matter. Information theory defines its quantities over a source and a channel: there is something sending, something receiving, and a distribution over what might have been sent. Those roles are formal and need nobody to occupy them —

reading a source as an observer is an error the theory does not commit, and it is worth not committing it here either. What is fixed is nonetheless a *distinguished pair of roles*, which is why entropy is a property of a source-and-receiver arrangement rather than of a thing. The restriction is of the same kind and arrived for the same reason: the field's problems were communication problems, and had two ends by construction. And graph theory, in its standard presentation, takes a vertex set first and defines edges as pairs drawn from it — which builds in the priority of relata over relations before any question about that priority has been asked. Neither field is careless. Each fixed an edge type or an order of construction because its own problems required one.

Which yields the deflationary version of this whole paper. *If everything is a graph, then the minimum structure required to be a graph is the least structure this account may retain.* That structure is not something this paper discovers: it has been written down in every textbook of the subject for a century, as relata and relations and nothing else. What is being done here is not deriving it but *taking it seriously* — reading a definition that was built as a formalism as though it were an answer to the largest question, and then checking that nothing else has to be added. The removals above are that check. That is the entire contribution, and it is smaller than it sounds while being, if the claim holds, sufficient.

One qualification even there. The textbook definition takes a vertex set as given and edges as pairs over it, which makes relata prior — and it does so for a good reason. *Graph theory's objects do not need to exist.* They are built, and building takes an order: you must have the vertices in hand before you can say which pairs of them are edges. The priority is a feature of *construction*, and construction is what mathematics does.

Nothing here is constructed. What exists was not assembled in an order, and there is no stage at which the relata were available and the relations not yet drawn. So the priority does not merely go unneeded in this paper — it is *unavailable*, and for a reason already established: an order of construction is a sequence, and there is no time here for one to run in. A definition that builds cannot be read as a description of something that was never built without carrying its scaffolding across.

Graph theory is not over-specified for its own purposes; it is exactly specified for constructing objects. What does not transfer is the scaffolding, and noticing that it is scaffolding is most of the work of taking the definition seriously.

That is also why the closest neighbour is the one furthest from physics. Nāgārjuna (1995) makes the claim in a form that does not require an observer edge — nothing needs to be looking, measuring or theorising for a thing to be nothing over and above what it stands in — which is the property the others lack, and which only became the discriminating one once the edge type was seen to be removable. That reading is contested, and the stronger form — that there is no observer edge anywhere in the account — is not one the secondary literature licenses: conceptual or nominal dependence is standardly the third and strongest of the dependence relations at issue, and on that reading the carving is ours. What the comparison needs is only the weaker claim, that the position is *statable* without the edge, which both readings allow.

There remains the fact that this is simple enough to state in a sentence and appears not to have been stated, which invites the reaction that it must therefore be either trivial or wrong. The account has its own explanation for that pairing and it is in the companion on assessment, so a short one will do here, and the short one is structural rather than a remark about incentives. *People do not study everything. They study something.* A field is constituted by its restriction — what it attends to and what it sets aside — and a field with no restriction is not a field. So the question this paper asks, which has to be put at maximum scope or it is not the same question, belongs to no discipline: there is no subject matter of which it is the subject matter, and no method calibrated for it, because every method is calibrated to a something.

That is why the general form goes unwritten while every local version gets written well.

And the mechanism has a worked case in this paper already, which makes the explanation

checkable instead of merely available. The gap between double entry being in daily use and the group of differences being written down is about three and a half centuries — Pacioli in 1494 to Hamilton in 1837, on Ellerman's own attribution — on top of the two centuries between the practice and Pacioli's description of it. That is not a hard construction: it is how the integers are built from the naturals.

The reason it went unwritten is *not* that nobody looked. Mathematicians did look — Ellerman names De Morgan in 1869 and Cayley, who published a pamphlet on double-entry bookkeeping in 1894 — and did not make the ordered-pairs identification. So the case is weaker than a clean two-restriction gap and is still a case: a trivial structure in continuous use by thousands of people, examined by capable people from the other side, went unformalised anyway. Ellerman adds the part that bears on the mechanism, which is that placing the identification took him three decades to publish in an accounting journal. The restriction is real; what it obstructed was the crossing rather than the looking.

The parallel is in the sociology and not in the content, and the difference matters. What double entry says is not what this paper says. What the case establishes is that *a thing can be simple, in constant use, and unwritten for centuries, for no reason except that no field owned it* — which is the mechanism this section claims, evidenced somewhere it can be dated. Whether this paper's claim is another instance of it is not something the case decides, and a reader who thinks the sentence is trivial and wrong rather than trivial and unwritten is disagreeing with the claim, which is the right place to disagree with it. It also explains, without needing a separate reason, why such statements tend to arrive from someone crossing between domains: crossing is what asking at maximum scope looks like from inside an arrangement where every position is a restricted one.

The claim of absence is held loosely, and more loosely than the rest of the paper, since nobody could be confident of having searched exhaustively. It is an invitation: if this has been written down as a claim about what exists, rather than as a formalism or as a thesis within one field, the right response to this paper is the reference.

What is contributed is accordingly not the thesis but its *position*: that it is enough on its own; that time, cause, rule, an observer, and any axiom about where distinctions may be made from are not required for it; and that an account of how things change can be built on top without folding any of them back in. The companion papers are that account, and their first move is to assume this one.

11 Falsifiability

Three commitments. The first is what the paper claims; the second is what the claim predicts and is the easiest to test; the third is a claim the paper makes about its own method.

They are not the same kind of claim, and calling all three falsifiable flattens a difference that matters. The first is *conceptual*, and is assessed the way claims of its kind are assessed: by whether it coheres, what it costs, what it forces elsewhere, and whether a rival buys more with less. No measurement settles it, and presenting it as though one might would be a borrowed credential. The second and third are ordinary empirical claims — about the history of formalisms, and about a method's yield — and they carry what risk is carried here. *The groundings carry the rest.* Where the companions put the claim to work in physics, in cosmology and in mind, it stops being a discipline on an account and becomes a commitment that can fail against observation; that is where the exposure sits, and it is deliberate that it sits there rather than here. Saying so is not a retreat from falsifiability. It is declining to claim a kind of testability this paper does not have, while naming the two places it does.

1. [SHAPE] **To be is to be related.** A claim about this instance and not about every possible one. Fails on the exhibition of an extant something standing in no relation whatever — not

merely one whose relations are unknown or unmeasured, which is the ordinary condition of most things, but one for which the claim that it stands in any relation at all is false. The burden is entirely on that side: no enumeration establishes the claim, since the domain does not terminate, and one instance would bound it.

An objection to this commitment has to be met, and meeting it says what kind of claim the proposition is. The objection runs: the paragraph above holds that the description of an unrelated existent *picks out nothing*, so the falsifying case is one in which there is no fact to report, and a falsifier nothing could answer to is not a falsifier. The first half is granted. The conclusion is not, because it assumes that a falsifier which cannot be *coherently* met is thereby not a falsifier, and the distinction those two run together is the one this whole account is about.

Neither bare verdict is available here, and reaching for one is where this goes wrong. Calling the commitment falsifiable, or calling it unfalsifiable, both import a frame in which a counterexample would be a *coherent* state of affairs — describable, then looked for. The reader supplies that frame at no cost, having no reason to suspect it is doing work, and it is the whole of the difficulty: the distinction this commitment turns on sits *below* the grain at which falsifiability is ordinarily argued, so a reader is not disagreeing about it but not resolving it. Said in full, then, and without the word that does not fit it: the commitment admits a determinate attempt with a determinate outcome, and that outcome cannot be a coherent counterexample. Calling this falsifiability borrows the credential the opening of this section declined; calling it unfalsifiability reports only that no measurement settles it, which was never in dispute. *The challenge this claim takes is conceptual* — an intelligible counterexample, or a rival account with lower cost at equal or better coverage. Drop the qualification and the sentence reverts to whichever verdict the reader already had.

A claim with no content has no attemptable counterexample at all: put one to a vocabulary and you get another expression in the vocabulary, which is not a failure of anything. Put one to this claim and the attempt has a determinate outcome — it decoheres, in the companion's exact sense, returning nothing with the whole of the frame as residual. That outcome is checkable, and it is distinguishable by inspection from the vocabulary case: *I tried and got nothing back* is a different result from *there was nothing to try.*

What the objection has found is not that the claim risks nothing. It is that the claim is at maximum scope, so the only thing that could refute it is the one thing the claim says cannot be assembled — and a question whose refutation would have to be incoherent is the most a question of this shape can be, not the least.

2. [SHAPE] **Compression moves toward graph form.** A consequence of the first and easier to test. Fails if a field, pushing its own formalism toward the shortest account that still suffices, moves *away* from relata-and-relations — adopting under compression pressure a primitive that is neither, and *keeping* it — carrying it in the formalism the field teaches and computes with rather than in a reduction nobody uses. What is asked is retention and not irreducibility, because graphs are expressive enough that almost any candidate can be argued to reduce, and a test whose confirming door is shut by that fact must not have its refuting door bolted by it too.

   Read strictly, that is already failed, and the failure is visible in this paper's own worked example. *The complete graph on a million vertices* keeps an arity and a count, and neither is a relatum or a relation; every confirming case above keeps something similar — amplitudes, irrep labels, branch lengths, edge weights — each taught and computed with. So the commitment cannot be about whether a formalism carries *only* relata and relations, since none does and none is likely to. What it can be about, and is here restricted to, is the *bearer of the field's distinctions*: whether what a field individuates and quantifies over under compression

pressure is relata and relations, with the rest carried as labels on them, or whether the individuated thing is something else. A field whose primitive bearer of distinction stopped being relata-and-relations would fail this. That is narrower, and it is what the evidence offered supports. It still lacks an operationalisation of *compression pressure*, which is the remaining hole. Expressiveness is not at issue in the other direction either: that a subject *can* be drawn as a graph is no evidence, and only the direction of movement under compression counts.

3. [SHAPE] **The method's yield is low.** The claim that the operation of declining an inherited restriction returns nothing on almost every attempt, which this paper leans on — it is what makes a rare result evidence of a discipline rather than of a machine for producing results. It is also a base rate over attempts nobody recorded, which is a defect and is stated as one. It fails if a record of such attempts, kept prospectively rather than reconstructed, shows the operation succeeding often; and this paper owes that record before the claim should be granted.

One objection to the first of these has to be met, because it is the one a reader reaches for first. It runs: exhibiting a counterexample means observing it, observing relates it to the observer, so anything offered as unrelated is related by the act of being offered, and the falsifier cannot be met.

That slides from *relation* to *observation*, and the two are not the same. *Observation is not required for existence.* The tree falls in the empty forest and stands in relation throughout — to the ground it strikes, to the air it moves, to its own fibres as they part — and not one of those relations waits on anybody looking. Observation is a *kind* of relation, and a late and expensive one, since it requires a frame with the machinery to make a distinction; relation is the general case and requires nothing.

So the claim here is not that to be is to be perceived, which is a different thesis with a different name and is not held. What is claimed is that to be is to be *related*, and the falsifier asks for something standing in no relation whatever — not for something unobserved, of which there is a great deal. That makes the commitment harder to meet rather than easier, since a candidate cannot be rescued by pointing out that it is now related to whoever produced it: the question is whether it stood in any relation at all, and the fallen tree settles that without an observer anywhere.

Proposition 5.1 survives this unchanged. What merely is still cannot observe. It simply does not follow that it needs to be observed in order to be.

A second thing would count against the paper without falsifying the proposition, and is worth stating separately: if time, cause, or rule turned out not to be separable — if the account of change built on this one could not be constructed without folding one of them back into the base — then the restraint claimed in Section 8 would be a fiction and this paper would be an incomplete statement of the base rather than a minimal one.

References

Anil Arya, John C. Fellingham, Jonathan C. Glover, Douglas A. Schroeder, and Gilbert Strang. Inferring transactions from financial statements. *Contemporary Accounting Research*, 17(3):366–385, 2000.

Anil Arya, John C. Fellingham, and Douglas A. Schroeder. Estimating transactions given balance sheets and an income statement. *Issues in Accounting Education*, 15(3):393–411, 2000.

Gregory Bateson. *Steps to an Ecology of Mind.* Chandler, San Francisco, 1972.

Max Black. The identity of indiscernibles. *Mind*, 61(242):153–164, 1952.

David P. Ellerman. The mathematics of double entry bookkeeping. *Mathematics Magazine*, 58(4):226–233, 1985.

David Ellerman. On double-entry bookkeeping: the mathematical treatment. *Accounting Education*, 23(5):483–501, 2014.

Michael Esfeld. Quantum entanglement and a metaphysics of relations. *Studies in History and Philosophy of Modern Physics*, 35(4):601–617, 2004.

James Ladyman and Don Ross. *Every Thing Must Go: Metaphysics Naturalized*. Oxford University Press, Oxford, 2007.

Geoffrey A. Lee. The coming of age of double entry: the Giovanni Farolfi ledger of 1299–1300. *Accounting Historians Journal*, 4(2):79–96, 1977.

Nāgārjuna. *The Fundamental Wisdom of the Middle Way: Nāgārjuna's Mūlamadhyamakakārikā*. Translated by Jay L. Garfield. Oxford University Press, New York, 1995.

Luca Pacioli. *Summa de Arithmetica, Geometria, Proportioni et Proportionalità*. Venice, 1494. The bookkeeping treatise is the *Particularis de Computis et Scripturis*.

Jorma Rissanen. Modeling by shortest data description. *Automatica*, 14:465–471, 1978.

Alan Sangster. The genesis of double entry bookkeeping. *The Accounting Review*, 91(1):299–315, 2016. Published online 2015.

Simon Saunders. Physics and Leibniz's principles. In Katherine Brading and Elena Castellani, editors, *Symmetries in Physics: Philosophical Reflections*, pages 289–307. Cambridge University Press, Cambridge, 2003.

Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 623–656, 1948.

John A. Wheeler. Information, physics, quantum: The search for links. In Wojciech H. Zurek, editor, *Complexity, Entropy, and the Physics of Information*, pages 3–28. Addison-Wesley, Redwood City, 1990.

Donald C. Williams. On the elements of being: I. *The Review of Metaphysics*, 7(1):3–18, 1953.

Basil S. Yamey. Scientific bookkeeping and the rise of capitalism. *The Economic History Review*, 1(2/3):99–113, 1949.